

Churn Analysis and Customer Intelligence

Churn Analysis

- Connect SQL database to Python – pandas and sqlite3
- Data import using SQL query in Python – pandas and sqlite3
- Data Cleaning – numpy and pandas
- data types, rename cols, select specific cols, handle missing/null values
- Feature Engineering – numpy and pandas
- Create new calculated cols, data transformation, use filters
- Data Analysis – numpy and pandas
- EDA - aggregation, group by, pivot table
- Data Visualization – matplotlib and seaborn

Churn Definition by Business

Churn analysis is the process of figuring out why customers stop doing business with you.

For this, we only need to look at three things:

- "Who": Identifying which customers left or will leave in future
- "Why": Analyzing their behavior before they left.
- "When": Finding the "danger zone" – point when a user leave

Executive Summary

The analysis was conducted on **21 customers** by integrating customer, subscription, and support datasets. The project focused on understanding customer churn, identifying high-risk customers, and recommending strategies to improve customer retention.

KPI Insights

1. Churn Rate

Value: **28.57%**

Insight

- Nearly **3 out of every 10 customers have churned**, indicating a relatively high customer loss rate.
- This suggests that customer retention should be a key business priority.

2. Retention Rate

Value: **71.43%**

Insight

- Approximately **7 out of every 10 customers remain active**.
- Although the majority of customers are retained, improving retention by even a small percentage could significantly increase long-term revenue.

3. Churn Rate by Plan Type

Plan	Churn Rate
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Basic	60.00%
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Standard	22.22%
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Premium	14.29%
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Insight

- Basic plan customers are the most likely to churn**, with a churn rate of **60%**.

- The **Premium plan has the lowest churn (14.29%)**, indicating stronger customer loyalty.
- Customers receiving more features or value appear less likely to leave.

Business Recommendation

- Improve the benefits of the Basic plan.
 - Offer upgrade incentives from Basic to Standard or Premium.
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4. Average Revenue Per User (ARPU)

Value: 18.85

Insight

- Each customer contributes an average of **18.85 units of monthly revenue**.
 - Increasing customer retention directly increases recurring revenue.
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5. Average Customer Tenure

Value: 1495.86 days

Insight

- Customers remain subscribed for an average of **approximately 4.1 years**, indicating good long-term loyalty among retained customers.
 - Long-tenure customers are valuable assets and should receive loyalty rewards.
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6. Revenue at Risk

Value: 73.94

Insight

- Churned customers account for **73.94 units of monthly revenue at risk**.
 - Reducing churn would directly recover recurring revenue and improve profitability.
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7. Escalation Rate

Value: 19.05%

Insight

- Nearly **1 in every 5 customers** required complaint escalation.
 - High escalation rates often indicate unresolved service issues and can negatively impact customer satisfaction.
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8. Average Complaints per User

Value: 0.43

Insight

- Customers filed fewer than **one complaint on average**, suggesting that complaint frequency is relatively low.
 - However, even a small number of complaints can significantly influence churn if not resolved effectively.
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9. Correlation Between Escalation and Churn

Value: 0.77

Insight

- The correlation coefficient of **0.77** indicates a **strong positive relationship** between complaint escalations and customer churn.
- Customers with escalated issues are much more likely to cancel their subscriptions.

Business Recommendation

- Prioritize faster resolution of escalated cases.
 - Implement proactive follow-up after escalations to improve customer satisfaction.
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Overall Business Findings

- **28.57% of customers have churned**, highlighting the need for stronger retention initiatives.
- The **Basic plan is the highest-risk segment**, with a **60% churn rate**, while Premium customers demonstrate much stronger loyalty.

- **Premium plan customers exhibit the lowest churn (14.29%)**, suggesting that perceived value plays a key role in retention.
 - Customers remain subscribed for an average of **1495.86 days**, showing that retained customers generate long-term value.
 - The business currently risks losing **73.94 units of recurring revenue** due to customer churn.
 - **19.05% of customers experienced complaint escalations**, and the strong **0.77 correlation** between escalations and churn indicates that poor support experiences are a major driver of customer loss.
 - An **ARPU of 18.85** highlights the importance of retaining existing customers, as recurring revenue depends on long-term subscriptions.
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Final Recommendations

1. Focus retention campaigns on **Basic plan customers**, where churn is highest.
 2. Strengthen customer support by reducing complaint escalation rates and improving resolution times.
 3. Identify customers with **high churn scores** and engage them proactively before renewal.
 4. Introduce loyalty rewards and personalized offers for long-tenure customers.
 5. Encourage Basic plan users to upgrade to Standard or Premium plans by highlighting additional benefits and value.
 6. Continuously monitor churn, complaint trends, and revenue at risk through dashboards to support data-driven decision-making.
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